

# MINGGANG LI

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## EDUCATION

### University of California, Berkeley

B.A. in Computer Science, Applied Mathematics, Statistics

Expected May 2028

GPA: 3.9/4.0

**Relevant Coursework:** Distributed Systems, Operating Systems, Algorithms, Machine Learning, Artificial Intelligence, Data Structures, Computer Architecture, Linear Algebra, Probability Theory

## EXPERIENCE

### Visa

*Software Engineer Intern, Payment Security and Identity Applications*

May 2026 - Aug 2026

*Foster City, CA*

- Building production backend services for **payment authentication** and **identity verification** supporting large-scale transaction workflows.
- Designing **distributed services** that handle high-throughput requests under strict reliability and latency SLAs.
- Contributing to **core payment security pipelines** and internal APIs powering **fraud detection** and credential protection alongside senior engineers.

### NASA Ames Research Center

*Machine Learning Engineer Intern*

Jan 2026 - May 2026

*Mountain View, CA*

- Engineered scalable training infrastructure on **Berkeley Savio HPC**, orchestrating **Slurm** job arrays with **PyTorch DDP** across **4-GPU** nodes to train transformer encoders for the **AI for Space Life Sciences (AI4SL)** group.
- Built **streaming preprocessing pipelines** that ingest large-scale **transcriptomic** matrices directly from **S3** into normalized Parquet shards, supporting reproducible multi-variant experiments without local disk staging.
- Benchmarked **linear-attention transformer** encoders and prototyped a **Mixture-of-Experts** routing layer over frozen domain experts, evaluated zero-shot on **NASA OSDR** spaceflight data to inform agency-wide **foundation model** system design.

### Berkeley Artificial Intelligence Research (BAIR) Lab

*Machine Learning Researcher*

Oct 2025 - May 2026

*Berkeley, CA*

- Developing **NLP tools** for early neurodegenerative disease detection under **Prof. Gopala Anumanchipalli**, building **semantic evaluation pipelines** that quantify narrative coherence and lexical diversity in patient speech.
- Implementing **transformer-based models** in **PyTorch** to extract semantic features from image descriptions; optimizing inference for low-latency clinical deployment.
- Engineered **content unit extraction algorithms** that automate cognitive decline tracking, replacing manual rater workflows with reproducible quantitative metrics.

### Gilead Sciences

*Machine Learning Engineer Intern*

Aug 2025 - Dec 2025

*Foster City, CA*

- Architected **Databricks ETL pipelines** that transformed 1000+ raw documents into structured **Snowflake** tables; project earned the "**Best Methodology: Large Language Modeling**" award for its Gen-AI pipeline design.
- Engineered a **parallel ensemble inference system** (Claude, Llama, FinBERT) with zero-shot prompting and score aggregation to automate high-throughput financial signal extraction from Wall Street reports.
- Delivered quantitative insights to executive leadership by applying **causal inference** models in **R** to correlate extracted text signals with market volatility.

## HONORS

**USACO Platinum Division** (Top 2% of competitive programmers nationally)

Jan 2023

## TECHNICAL SKILLS

### Languages

Python, Java, C/C++, SQL, R, JavaScript, HTML/CSS, **Assembly (RISC-V)**

### Frameworks

**PyTorch**, TensorFlow, Scikit-learn, **React**, **Node.js**, Spark, Firebase

### Cloud & Data

AWS, GCP, **Databricks**, **Snowflake**, Parquet, W&B

### Infra & Tools

**Slurm**, **Savio HPC**, **Docker**, Kubernetes, **Git**, **Linux**